Yuqi Jia

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Education

Duke University — Durham, US
Ph.D. student in ECE — Aug. 2022 – Present
Advisor: **Prof. Neil Gong**

Duke University — Durham, US
M.S. in Electrical and Computer Engineering — Aug. 2022 – Sep. 2025

Shanghai Jiao Tong University — Shanghai, China
B.S. in Computer Science — Sept. 2018 – June 2022
Member of **ACM Honors Class**, an elite CS program for top five percent of students.

Research Interests

LLM Agents, Reinforcement Learning, Post Training
AI Security, Trustworthy AI

Research Experience

Research Scientist Intern, TikTok — May. 2026 – Oct. 2026
Research topic: Agent Foundation Model

- Built an **open agent foundation model** with a simplified **three-stage post-training** recipe (two SFT stages and RL), plus a **data pipeline** synthesizing difficulty-enhanced reasoning problems and automatically verifiable tool-use tasks.
- Resulting **35B** and **122B** models match or exceed frontier models on six reasoning and agentic benchmarks. Paper under internal review.

Research Scientist Intern, ByteDance — May. 2025 – Sep. 2025
Research topic: RL-based Redteaming Attack against Multi-agent Systems

- Led a project on **automated red-teaming of multi-agent LLM systems**, fine-tuning attacker LLMs with an RL framework (**DAPO**) to generate prompts that extract sub-agents' protected system prompts under **black-box** access. paper

Research Scientist Intern, ByteDance — June. 2024 – August. 2024
Research topic: Prompt Injection Attacks

- Surveyed and empirically evaluated existing **prompt-injection attacks and defenses** across **1,000+ agents** deployed on the company's platform.
- Designed a precise and efficient metric for detecting system-prompt leakage.

Undergraduate researcher, Mila-Quebec AI Institute

advised by **Prof. Jian Tang** May. 2021 – Nov. 2021
Research topic: Graph Structure Learning

Undergraduate Researcher, ThinkLab, Shanghai Jiao Tong University
advised by **Prof. Junchi Yan** Aug. 2020 – July 2021
Research topic: Adversarial Attack

Selected Publications

Black-Box Red-Teaming of Multi-Agent Systems via Reinforcement Learning. PDF
Y Jia, H Zhang, C Liu, J Liang, and X Chen

ContextLeak: Exfiltrating LLM Agent Context via Malicious Tools. PDF
Y Jia, R Wang, P Li, Y Hu, P Li, and NZ Gong

Visual Contrastive Self-Distillation. PDF
Y Liang, Y Tian, Y Li, **Y Jia**, F Huang, T Zhou, and D Fu

ObliInjection: Order-Oblivious Prompt Injection Attack to LLM Agents with Multi-source Data. PDF
R Wang, **Y Jia**, and NZ Gong
Network and Distributed System Security (NDSS) Symposium 2026.

MalTool: Malicious Tool Attacks on LLM Agents. PDF
Y Hu, **Y Jia**, M Li, D Song, and NZ Gong

PromptLocate: Localizing Prompt Injection Attacks. PDF
Y Jia, Y Liu, Z Shao, J Jia, and NZ Gong
IEEE Symposium on Security and Privacy 2026.

Competitive Advantage Attacks to Decentralized Federated Learning. PDF
Y. Jia, M. Fang and N. Gong
Conference on Neural Information Processing Systems (NeurIPS) 2025.

Tracing Back the Malicious Clients in Poisoning Attacks to Federated Learning. PDF
Y. Jia, M. Fang, H. Liu, J. Zhang and N. Gong
Conference on Neural Information Processing Systems (NeurIPS) 2025.

DataSentinel: A Game-Theoretic Detection of Prompt Injection Attacks. PDF
Y Liu, **Y Jia**, J Jia, D Song and NZ Gong
IEEE Symposium on Security and Privacy 2025. **Distinguished Paper Award**

Evaluating Large Language Model based Personal Information Extraction and Countermeasures. PDF
Y Liu, **Y Jia**, J Jia and NZ Gong

USENIX Security Symposium 2025.

Formalizing and benchmarking prompt injection attacks and defenses. PDF
Y. Liu, **Y. Jia**, R. Geng, J. Jia, N. Gong
USENIX Security Symposium 2024.

Unlocking the Potential of Federated Learning: The Symphony of Dataset Distillation via Deep Generative Latents. PDF
Y. Jia*, S. Vahidian*, J. Sun, J. Zhang, V. Kungurtsev, N. Gong and Y. Chen
European Conference on Computer Vision (ECCV) 2024.

Invited Talks

Berkeley Security Seminar, University of California, Berkeley link — May 2026
CISC 849, University of Delaware, hosted by Prof. Xing Gao — Mar. 2026

Honors and Scholarships

Scholarship
Zhiyuan Honorary Scholarship (**Top 5%** in SJTU) 2018-2021
Excellent School-level Scholarship (**Top 3%** in SJTU) 2018-2021

Competition
Silver Medal (**rank 1st**), The 33rd Chinese Mathematical Olympiad(CMO) 2017
Honorable Mention, Interdisciplinary Contest In Modeling(ICM) 2020
First Prize, Mathematics competition of Chinese College Students(CMC) 2019

Academic Activities

Conference Reviewer
USENIX Security 2026; ACM CCS 2026, 2025; NeurIPS 2026; ICLR 2026,2025; ICML 2026

Teaching Experience

Teaching Assistant, Duke University Spring 2025
Generative AI: Foundations, Applications, and Safety, ECE590

Teaching Assistant, Duke University Fall 2024
Natural Language Processing, ECE684

Teaching Assistant, Shanghai Jiao Tong University Fall 2020
Mathematical Logic, CS301-1

Skills

Programming Skills: Python, C/C++, PyTorch
Language proficiency: Chinese(Native), English.